

Solution:

```
import java.util.Scanner;

public class Quiz1{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the lottery of the day (2 digits): ");
        int lottery = sc.nextInt();
        System.out.print("Guess the lottery number (2 digits): ");
        int guess = sc.nextInt();

        int lotteryDigit1 = lottery / 10;
        int lotteryDigit2 = lottery % 10;
        int guessDigit1 = guess / 10;
        int guessDigit2 = guess % 10;

        if (guess == lottery)
            System.out.println("Exact match: you win $10,000");
        else if (guessDigit2 == lotteryDigit1 && guessDigit1 == lotteryDigit2)
            System.out.println("Match all digits: you win $3,000");
        else if (guessDigit1 == lotteryDigit1
            || guessDigit1 == lotteryDigit2
            || guessDigit2 == lotteryDigit1
            || guessDigit2 == lotteryDigit2)
            System.out.println("Match one digit: you win $1,000");
        else
            System.out.println("Sorry, no match");
    }
}
```

Question - 1: [14 points]

The university mathematics club is hosting an event called **Fun with Factorial!**.

As a part of that event, they are choosing N integers of which they want to find the factorials as well as the maximum factorial value.

Factorial of a number: Multiplication from 1 till that number. Ex: $4! = 1*2*3*4 = 24$.

Write a java program to take the N number of integers as user inputs from which their factorial and maximum factorial is to be found.

Note: You cannot use Arrays. Assume the user will not give a negative number.

Sample Input	Sample Output
Enter the number of integers (N): 4 Enter integer 1: 7 Enter integer 2: 2 Enter integer 3: 5 Enter integer 4: 3	Factorial of 7 is: 5040 Factorial of 2 is: 2 Factorial of 5 is: 120 Factorial of 3 is: 6 Maximum factorial value: 5040
Enter the number of integers (N): 5 Enter integer 1: 3 Enter integer 2: 5 Enter integer 3: 4 Enter integer 4: 2 Enter integer 5: 6	Factorial of 3 is: 6 Factorial of 5 is: 120 Factorial of 4 is: 24 Factorial of 2 is: 2 Factorial of 6 is: 720 Maximum factorial value: 720

Question - 2: [6 points]

Write the outputs (5 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

	public class Tracing_Quiz2 {	Output:
2	public static void main(String[] args) {	
3	int a = 4, b = 9, c = 1;	
4	double d = 2.5;	1.
5	while (true) {	
6	++a;	2.
7	double val;	
8	if (a == 7) {	3.
9	continue;	
10	}	4.
11	if (a % 3 == 0) {	
12	b = (b / a) + c++;	5.
13	System.out.println(b); }	
14	else {	
15	val = (b / a) + c * ((int)d);	
16	System.out.println(val); }	
17	c -= - a + 1;	
18	if (a == 10) {	
19	break;	
20	}	
21	}	
22	}	
23	}	

BRAC UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
QUIZ II
CSE110: PROGRAMMING LANGUAGE I

NAME:	ID:	SECTION:
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Question - 2: [6 points]

Write the outputs (5 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

		Output:
1	public class Tracing_Quiz2 {	
2	public static void main(String[] args) {	
3	int a = 3, b = 12, c = 2;	
4	double d = 3.5;	1.
5	while (true) {	
6	++a;	2.
7	double val;	
8	if (a == 7) {	3.
9	continue;	
10	}	4.
11	if (a % 3 == 0) {	
12	b = (b / a) + c++;	5.
13	System.out.println(b); }	
14	else {	
15	val = (b / a) + c * ((int)d);	
16	System.out.println(val); }	
17	c -= -a + 1;	
18	if (a == 9) {	
19	break;	
20	}	
21	}	
22	}	
23	}	

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Question - 2: [6 points]

Write the outputs (5 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

1	public class Tracing_Quiz2 {	Output:
2	public static void main(String[] args) {	
3	int a = 5, b = 20, c = 3;	1.
4	double d = 1.8;	
5	while (true) {	
6	++a;	
7	double val;	
8	if (a == 7) {	2.
9	continue;	3.
10	}	4.
11	if (a % 3 == 0) {	5.
12	b = (b / a) + c++;	
13	System.out.println(b); }	
14	else {	
15	val = (b / a) + c * ((int)d);	
16	System.out.println(val); }	
17	c -= - a + 1;	
18	if (a == 11) {	
19	break;	
20	}	
21	}	
22	}	
23	}	

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Write a java program to take the N number of integers as user inputs from which their factorial and maximum factorial is to be found.

Note: You cannot use Arrays. Assume the user will not give a negative number.

Sample Input	Sample Output
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Enter the number of integers (N): 5 Enter integer 1: 3 Enter integer 2: 5 Enter integer 3: 4 Enter integer 4: 2 Enter integer 5: 6	Factorial of 3 is: 6 Factorial of 5 is: 120 Factorial of 4 is: 24 Factorial of 2 is: 2 Factorial of 6 is: 720 Maximum factorial value: 720

Question - 2: [6 points]

Write the outputs (5 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

1	public class Tracing_ Quiz2 {	Output:
2	public static void main(String[] args) {	
3	int a = 1, b = 15, c = 4;	1. 2. 3. 4. 5.
4	double d = 2.2;	
5	while (true) {	
6	++a;	
7	double val;	
8	if (a == 7) {	
9	continue;	
10	}	
11	if (a % 3 == 0) {	
12	b = (b / a) + c++;	
13	System.out.println(b); }	
14	else {	
15	val = (b / a) + c * ((int)d);	
16	System.out.println(val); }	
17	c -= - a + 1;	
18	if (a == 6) {	
19	break;	
20	}	
21	}	
22	}	
23	}	

Tracing:

Set 1

3.0

6

22.0

18

55.0

Set 2:

9.0

17.0

11

46.0

23

Set 3:

6

9.0

16

26.0

35.0

Set 4:

15.0

10

18.0

24.0

16

```
public class QUIZZCode {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter the number of integers (N): ");  
        int N = sc.nextInt();  
        int maxFactorial = 0;  
        for (int i = 0; i < N; i++) {
```

```
            System.out.print("Enter integer " + i + ": ");  
            int num = sc.nextInt();  
            int factorial = 1;  
            for (int j = 1; j <= num; j++) {  
                factorial *= j;  
            }  
            System.out.println("Factorial of " + num + " is: " + factorial);  
            if (i == 0)  
            {  
                maxFactorial = factorial;  
            }  
            else if (factorial > maxFactorial) {  
                maxFactorial = factorial;  
            }  
        }  
        System.out.println("Maximum factorial value: " + maxFactorial);  
    }  
}
```

Question - 1: [20 points]

As a software engineer, you have been given an array of football player names and their jersey numbers, which are conjoined together in strings. You are now given the task to separate the names and the jersey numbers in **two different arrays**.

Note: Your code should work for arrays of any length. You cannot use any built-in method **except** `length()` and `charAt()`.

Sample Given Array	Sample Output
<code>String [] players = {"10 Messi", "Ronaldo7", "Beckham 7", "Andres8"};</code>	Names: [Messi, Ronaldo, Beckham, Andres] Jersey Numbers: [10, 7, 7, 8]
<code>String [] players = {"Haaland 9", "M-10-essi", "7Ronaldo", "And8res", "Beckham7"};</code>	Names: [Halland, Messi, Ronaldo, Andres, Beckham] Jersey Numbers: [9, 10, 7, 8, 7]

```
import java.util.Arrays;
public class Quiz3_Sec20 {
    public static void main(String[] args) {
        String [] players = {"10 Messi", "Ronaldo7", "Beckham 7", "Andres8"};
        String [] names = new String[players.length];
        String [] jersey_numbers = new String[players.length];
        for (int i =0; i<players.length; i++)
        {
            String temp1 = "";
            String temp2 = "";
            for (int j = 0; j<players[i].length(); j++)
            {
                if (players[i].charAt(j)>='0' && players[i].charAt(j)<='9')
                {
                    temp1 += players[i].charAt(j);
                }
                else if ((players[i].charAt(j)>='A' && players[i].charAt(j)<='Z') ||
(players[i].charAt(j)>='a' && players[i].charAt(j)<='z'))
                {
                    temp2 += players[i].charAt(j);
                }
            }
            names[i] = temp2;
            jersey_numbers[i] = temp1;
        }
        System.out.println(Arrays.toString(names));
        System.out.println(Arrays.toString(jersey_numbers));
    }
}
```

write the outputs (9 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

	public class Quiz4 {	Output:
2	public static void main(String[] args) {	
3	System.out.println(method1 ("Derek", "The", "Delightful"));	
4	}	1.
5	public static boolean method1(String s1, String s2, String s3) {	
6	method2 (s1.length(), s2.length());	2.
7	return (s1 + s2).charAt(0) == s3.charAt(0);	
8	}	3.
9	public static void method2(int num1, int num2) {	
10	System.out.println(num1 / 2 + num2++);	4.
11	System.out.println(num2 - 10 / num1);	
12	System.out.println(method3 (num2));	5.
13	}	
14	public static int method3(int a)	6.
15	{	
16	if (a < 0) {	7.
17	return a + 2;	
18	}	8.
19	int b = --a + 5;	
20	int x = method3 (a) - 1;	9.
21	System.out.println(x + method4 (b));	
22	return a + x;	
23	}	
24	public static int method4(int c) {	
25	return c + 4;	
26	}	
27	}	

trace table and calculation, for which 1 mark will be allocated.

		Output:
1	public class Quiz4 {	1. 2. 3. 4. 5. 6. 7. 8. 9.
2	public static void main(String[] args) {	
3	System.out.println(method1 ("Steve", "The", "Saviour"));	
4	}	
5	public static boolean method1(String s1, String s2, String s3) {	
6	method2 (s1.length(), s2.length());	
7	return (s1 + s2).charAt(0) == s3.charAt(0);	
8	}	
9	public static void method2(int num1, int num2) {	
10	System.out.println(num1/ 2 + (++num2));	
11	System.out.println(num2 - 9 / num1);	
12	System.out.println(method3 (num2));	
13	}	
14	public static int method3(int a)	
15	{	
16	if (a < 0) {	
17	return a + 1;	
18	}	
19	int b = --a + 4;	
20	int x = method3 (a) - 2;	
21	System.out.println(x + method4 (b));	
22	return a + x;	
23	}	
24	public static int method4(int c) {	
25	return c + 3;	
26	}	
27	}	

trace table and calculation, for which 1 mark will be allocated.

1	public class Quiz4 {	Output:
2	public static void main(String[] args) {	
3	System.out.println(method1 ("Holly", "The", "Wise"));	
4	}	
5	public static boolean method1(String s1, String s2, String s3) {	
6	method2 (s1.length(), s2.length());	
7	return (s1 + s2).charAt(0) == s3.charAt(0);	
8	}	
9	public static void method2(int num1, int num2) {	
10	System.out.println(num1 / 5 + (++num2));	
11	System.out.println(num2 - 14 / num1);	
12	System.out.println(method3 (num2));	
13	}	
14	public static int method3(int a)	
15	{	
16	if (a < 0) {	
17	return a + 5;	
18	}	
19	int b = --a + 1;	
20	int x = method3 (a) + 6;	
21	System.out.println(x + method4 (b));	
22	return a + x;	
23	}	
24	public static int method4(int c) {	
25	return c + 2;	
26	}	
27	}	

trace table and calculation, for which 1 mark will be allocated.

		Output:
1	public class Quiz4 {	
2	public static void main(String[] args) {	
3	System.out.println(method1 ("Mike", "The", "Master"));	
4	}	1.
5	public static boolean method1(String s1, String s2, String s3) {	
6	method2 (s1.length(), s2.length());	2.
7	return (s1 + s2).charAt(0) == s3.charAt(0);	
8	}	3.
9	public static void method2(int num1, int num2) {	
10	System.out.println(num1 / 7 + num2++);	4.
11	System.out.println(num2 - num1 / 6);	
12	System.out.println(method3 (num2));	5.
13	}	
14	public static int method3(int a)	6.
15	{	
16	if (a < 0) {	7.
17	return a - 1;	
18	}	8.
19	int b = --a + 6;	
20	int x = method3 (a) + 7;	9.
21	System.out.println(x + method4 (b));	
22	return a + x;	
23	}	
24	public static int method4(int c) {	
25	return c - 1;	
26	}	
27	}	

Set A:

1. 5
2. 2
3. 8
4. 7
5. 7
6. 8
7. 10
8. 1
9. True

Set B:

1. 6
2. 3
3. 4
4. 2
5. 1
6. 1
7. 2
8. -5
9. True

Question - 1: [20 points]

Suppose you want to develop a program to play the lottery. The program takes the lottery number of the day as user input (**assume it is a 2 digit number**), prompts the user to enter a **2 digit number as a guess**, and determines whether the user wins according to the following rules:

1. If the user's guess matches the lottery number in the **exact order**, the award is \$10,000.
2. If all digits in the user's guess **match all digits in the lottery number, but not in exact order**, the award is \$3,000.
3. If one digit of the user's guess **matches a digit** in the lottery number, the award is \$1,000.

Sample Input	Sample Output
Enter the lottery of the day (2 digits)!: 15 Guess the lottery number (2 digits): 15	Exact match: you win \$10,000
Enter the lottery of the day (2 digits)!: 15 Guess the lottery number (2 digits): 51	Match all digits but not exact order: you win \$3,000
Enter the lottery of the day (2 digits)!: 23 Guess the lottery number (2 digits): 34	Match one digit: you win \$1,000
Enter the lottery of the day (2 digits)!: 15 Guess the lottery number (2 digits): 44	Sorry: no match

Set C:

1. 5
2. 2
3. 12
4. 18
5. 25
6. 33
7. 42
8. 39
9. false

Set D:

1. 3
2. 4
3. 9
4. 16
5. 24
6. 33